Education, Child Marriage, and Heterogeneous Wage Returns among Women in Nigeria

Results

Hernando Grueso, Ritapriya Bandyopadhyay, Chris Desmond, Brendan Maughan-Brown, Lucie Cluver, William Rudgard

25/08/2026

Early marriage remains widespread in Northern Nigeria and may constrain women's economic gains from schooling. Using the 2018–19 Nigeria Living Standards Survey (n=2,387 wage earners nationally; n=932 in Northern Nigeria), we estimate Mincerian wage models with state fixed effects, Heckman selection correction, and causal forests.

Each additional year of schooling raises wages by ~15% in both samples, with convex returns. Child marriage (before 18) is associated with a 14.4% wage penalty nationally (12% prevalence) and 23.6% in Northern Nigeria (23% prevalence).

Nationally, the interaction is significant ($p<0.001$): child marriage reduces both wage levels and returns to education. In Northern Nigeria, child marriage operates as a pure level shift (interaction $p=0.459$).

Main Results

Whole Country (12% prevalence)

- Significant interaction ($p < 0.001$)
- Child marriage = level shift + reduces returns
- Returns: 7–9% (married) vs 4–14% (unmarried)
- Variable importance: 6.8%

Northern Nigeria (23% prevalence)

- No interaction ($p = 0.459$)
- Child marriage = pure level shift
- Returns stay ~15% regardless of marriage age
- Variable importance: only 4.9%

Policy Implications

1. Each year of schooling raises wages by ~15%, and child marriage lowers wage levels by ~15%. In Northern Nigeria, these effects are additive: policies must scale up schooling and delay marriage in parallel.
2. Northern Nigeria's child marriage prevalence (23%) and wage penalty (~15%) are roughly double the national averages.
3. In high-prevalence settings, the penalty likely reflects broad social constraints (fertility timing, mobility limits, norms), calling for community-wide and cash-plus approaches.

Descriptive Statistics

Table 1: Descriptive Statistics (Wage Earners)

Table 1: Descriptive Statistics – Wage Earners

Variable	National (N=2,387)	N. Nigeria (N=932)
Continuous: Mean (SD)		
Log annual wage	12.02 (1.42)	11.83 (1.60)
Years of schooling	14.42 (5.81)	13.06 (7.02)
Age	36.69 (11.03)	36.34 (10.85)
Experience	16.01 (12.13)	16.65 (11.91)
Binary (%)		
Child marriage (%)	11.7	23.1
Rural (%)	49.9	56.3
Literate (%)	86.1	78.9
Currently married (%)	61.0	70.7
Religion (%)		
Religion: Islam (%)	76.5	53.1
Religion: Christianity (%)	23.0	46.8
Religion: Other (%)	0.5	0.1
Father education (%)		
Father edu: None (%)	33.6	38.1
Father edu: Primary (%)	43.6	28.2
Father edu: Secondary+ (%)	22.8	33.7
Mother education (%)		
Mother edu: None (%)	45.1	51.2
Mother edu: Primary (%)	40.9	28.2
Mother edu: Secondary+ (%)	14.0	20.6
Sector (%)		
Sector: Urban (%)	50.1	43.7
Sector: Rural (%)	49.9	56.3
N	2,387	932

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Continuous variables: Mean (SD).

National Results

Table 2: Sample Construction – National

Step	Description	N	N_dropped
0	Full Whole Country dataset	27,488	0
1	Women only	27,488	0
2	Age 18-64 (working age)	27,488	0
3	Non-missing key variables	26,450	1,038
4	Workers with valid wage data	2,387	24,063

Table 2: National – Quadratic Mincer

Table 3: National – Quadratic Mincer

	Basic	+ Controls	+ State FE	+ Interaction	Heckman
Years of Schooling	0.116***	0.111***	0.089***	0.082***	0.077***
	(0.006)	(0.006)	(0.007)	(0.008)	(0.003)
Years of Schooling²	0.007***	0.007***	0.008***	0.009***	0.009***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
Experience	0.042***	0.043***	0.040***	0.040***	0.020***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Experience²	−0.001***	−0.001***	−0.001***	−0.001***	−0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Child Marriage	−0.314***	−0.263***	−0.078	−0.144	
	(0.081)	(0.081)	(0.084)	(0.091)	
Years × Child Marriage				0.022*	
				(0.011)	
Rural		−0.261***	−0.145***	−0.142***	
		(0.044)	(0.048)	(0.048)	
Inv. Mills Ratio ( )					0.822**
					(0.361)
Currently Married		−0.098**	−0.035	−0.035	
		(0.047)	(0.046)	(0.046)	
Observations	2387	2387	2387	2387	26 432
R²	0.403	0.412	0.479	0.480	
Adj. R²	0.402	0.411	0.470	0.470	
Residual Std. Error					1.222
State FE	No	No	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Interaction	No	No	No	Yes	No
Selection Correction	No	No	No	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. School-

Table 4: National – Linear Specification

	Basic	+ Controls	+ State FE	+ Interaction
Years of Schooling	0.169***	0.163***	0.152***	0.156***
	(0.005)	(0.005)	(0.005)	(0.006)
Experience	0.047***	0.046***	0.044***	0.045***
	(0.002)	(0.002)	(0.002)	(0.002)
Child Marriage	−0.120	−0.087	0.031	0.314**
	(0.083)	(0.083)	(0.087)	(0.154)
Years × Child Marriage				−0.025**
				(0.011)
Rural		−0.270***	−0.181***	−0.182***
		(0.046)	(0.050)	(0.050)
State FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Interaction	No	No	No	Yes
Currently Married		0.087*	0.140***	0.135***
		(0.046)	(0.045)	(0.046)
Observations	2387	2387	2387	2387
R^2	0.362	0.372	0.433	0.434
Adj. R^2	0.361	0.371	0.423	0.424

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied.

Table 5: Returns by Education Level – National

	By Level
Primary	0.205
	(0.216)
Secondary	0.309**
	(0.122)
Higher Ed	1.558***
	(0.122)
Experience	0.041***
	(0.002)
Experience Sq.	−0.001***
	(0.000)
Child Marriage	−0.167*
	(0.093)
N	2387
R2	0.369

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note:
Omitted: No education. State FE, rural, marital controls included.

Table 5: Returns to Literacy – National

Table 6: Returns to Literacy – National

	Literacy
Literate	1.159***
	(0.089)
Experience	0.029***
	(0.002)
Experience Sq.	–0.001***
	(0.000)
Child Marriage	–0.242**
	(0.095)
N	2387
R2	0.323

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred.

Table 7: Probit: Pr(Wage Employment) – National

	Pr(Wage Work)
Years of Schooling	0.076***
	(0.003)
Years Sq.	0.009***
	(0.000)
Experience	0.020***
	(0.001)
Experience Sq.	−0.001***
	(0.000)
Child Marriage	−0.123***
	(0.040)
Rural	−0.158***
	(0.029)
Observations	26 450
State FE	Yes
Parent Education	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE.

Table 8: Average Marginal Effects on Pr(Wage Work) – National

Variable	AME	SE	p
Child Marriage	-0.0153	0.0048	0.001
Experience	0.0025	0.0002	< 0.001
Experience Sq.	-0.0001	0.0000	< 0.001
Rural	-0.0209	0.0040	< 0.001
Years of Schooling	0.0098	0.0004	< 0.001
Years of Schooling Sq.	0.0011	0.0000	< 0.001

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. AME via marginaleffects package.

Table 8: Causal Forest – National

Average Return to Education

Table 9: CF: Average Treatment Effect – National

Term	Estimate	Std. Error	t stat	p value
Education Return (per year)	0.155	0.017	9.346	0

Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees.

Test for Heterogeneity

Table 10: CF Calibration Test – National

Term	Estimate	Std. Error	t stat	p value
Mean Forest Prediction	1.045	0.184	5.678	0.00
Differential Forest Prediction	0.668	0.355	1.880	0.03

Variable Importance

Table 11: CF Variable Importance – National

Variable	Importance
Experience	0.319
Mother's Education	0.230
Age	0.229
Father's Education	0.126
Child Marriage	0.064
Rural	0.032

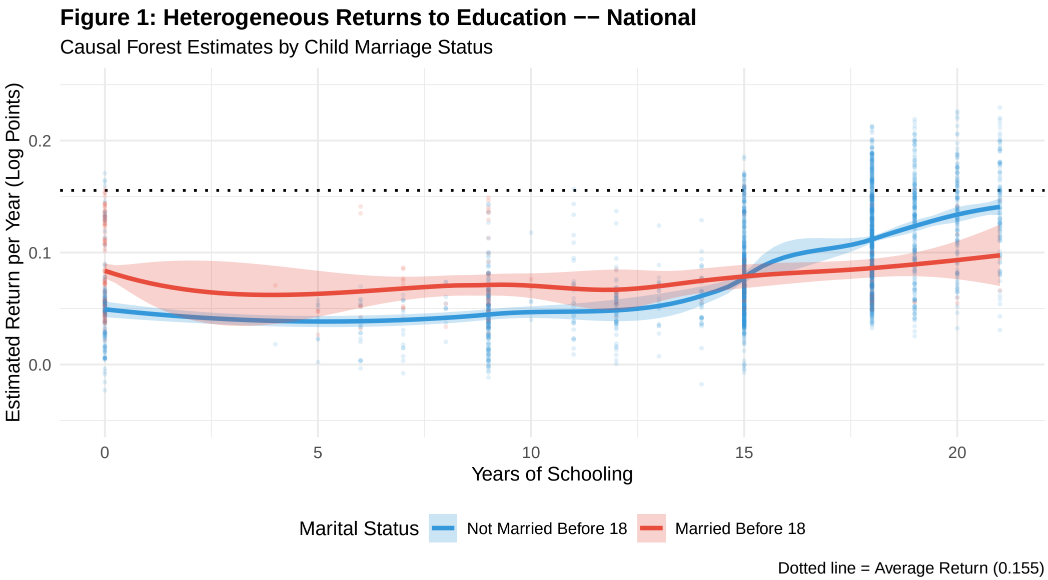


Figure 1: Heterogeneous Returns to Education –– National
Causal Forest Estimates by Child Marriage Status
Estimated Return per Year (Log Points)
0.2
0.1
0.0
0
5
10
15
20
Years of Schooling
Marital Status
Not Married Before 18
Married Before 18
Dotted line = Average Return (0.155)

Figure 2: Distribution of Returns – National

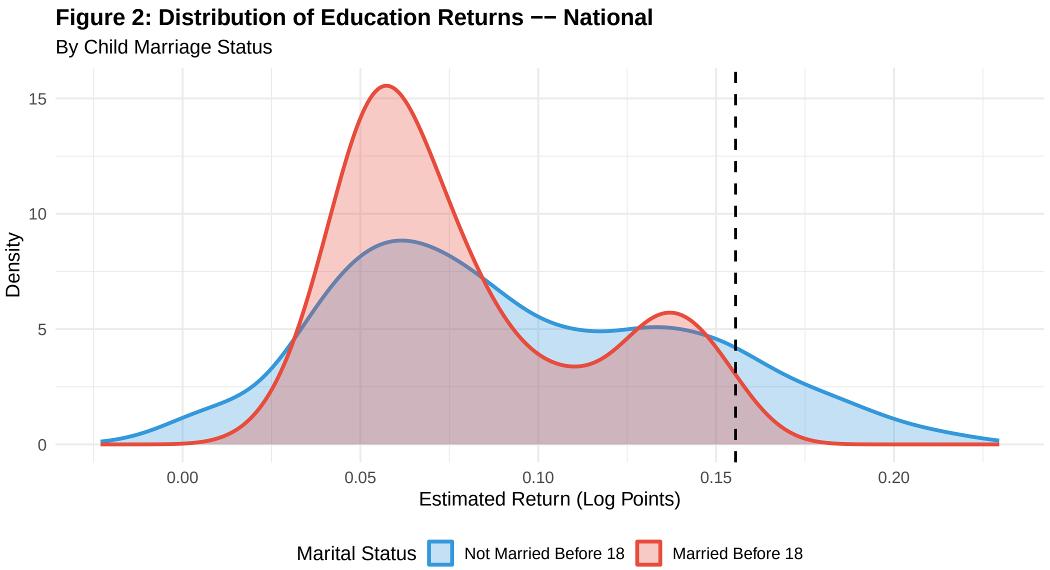
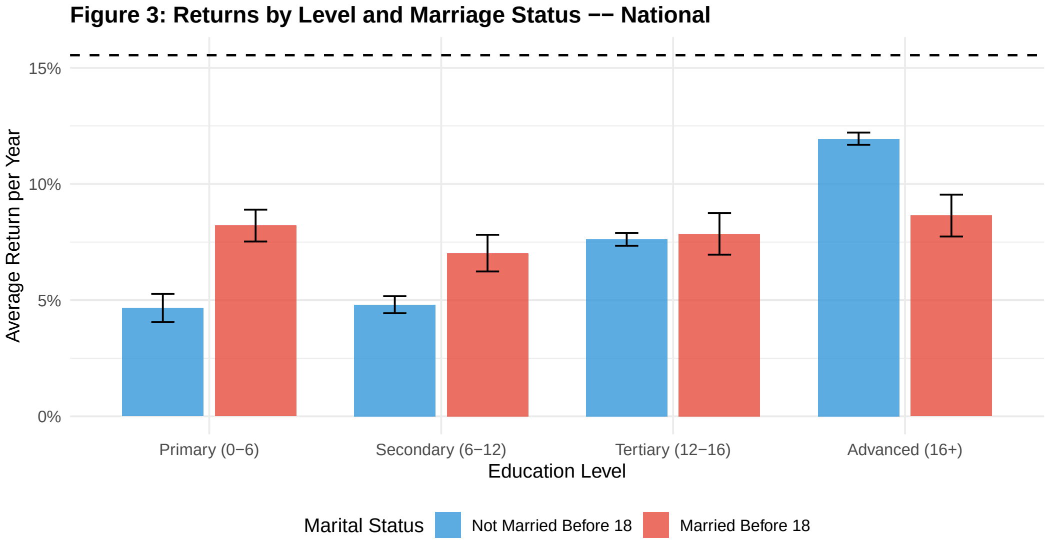


Figure 3: Returns by Level and Marriage – National



Dashed line = overall average. Error bars = 95% CIs.

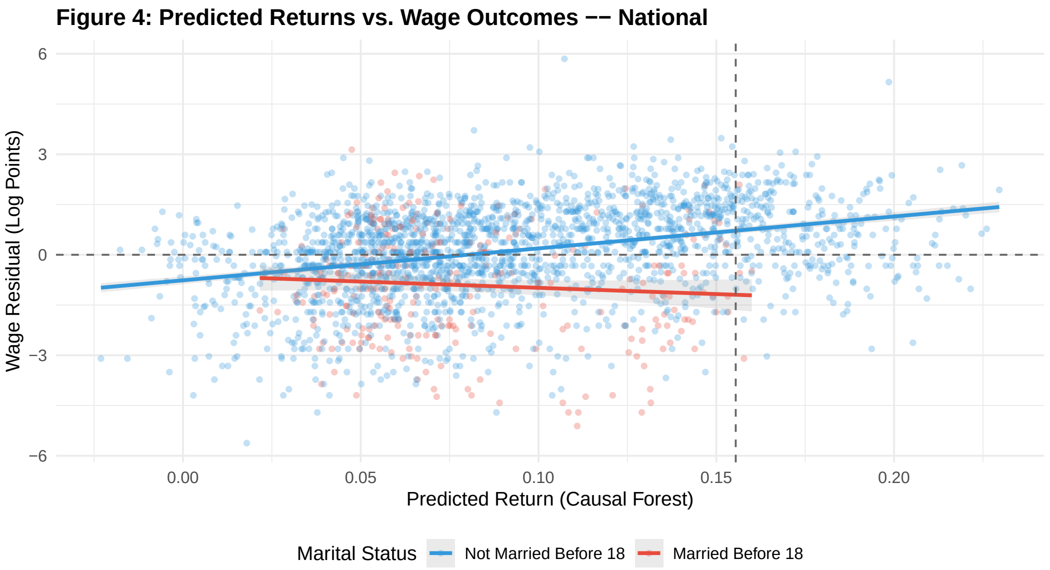


Figure 4: Predicted Returns vs. Wage Outcomes –– National
Wage Residual (Log Points)
6
3
0
−3
−6
0.00
0.05
0.10
0.15
0.20
Predicted Return (Causal Forest)
Marital Status
Not Married Before 18
Married Before 18

Northern Nigeria

Table 12: Sample Construction – Northern Nigeria

Step	Description	N	N_dropped
0	Full Northern Nigeria dataset	16,611	0
1	Women only	16,611	0
2	Age 18-64 (working age)	16,611	0
3	Non-missing key variables	16,140	471
4	Workers with valid wage data	932	15,208

Table 13: Northern Nigeria – Quadratic Mincer

	Basic	+ Controls	+ State FE	+ Interaction	Heckman
Years of Schooling	0.071***	0.066***	0.053***	0.043***	0.048***
	(0.011)	(0.011)	(0.011)	(0.013)	(0.005)
Years of Schooling²	0.012***	0.012***	0.011***	0.011***	0.009***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Experience	0.042***	0.041***	0.040***	0.039***	0.026***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)
Experience²	−0.001***	−0.001***	−0.001***	−0.001***	−0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Child Marriage	−0.262***	−0.237**	−0.150	−0.236**	
	(0.094)	(0.094)	(0.096)	(0.113)	
Years × Child Marriage				0.017	
				(0.012)	
Rural		−0.253***	−0.130*	−0.123*	
		(0.074)	(0.073)	(0.073)	
Inv. Mills Ratio ( )					0.217
					(0.420)
Currently Married		−0.187**	−0.123	−0.113	
		(0.080)	(0.078)	(0.079)	
Observations	932	932	932	932	16 130
R^2	0.540	0.548	0.605	0.606	
Adj. R^2	0.537	0.544	0.594	0.594	
Residual Std. Error					1.021
State FE	No	No	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Interaction	No	No	No	Yes	No
Selection Correction	No	No	No	No	Yes

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. School-

Table 14: Northern Nigeria – Linear Specification

	Basic	+ Controls	+ State FE	+ Interaction
Years of Schooling	0.172***	0.166***	0.149***	0.155***
	(0.007)	(0.007)	(0.007)	(0.009)
Experience	0.047***	0.046***	0.045***	0.045***
	(0.004)	(0.004)	(0.003)	(0.003)
Child Marriage	−0.142	−0.112	−0.044	0.148
	(0.102)	(0.101)	(0.103)	(0.170)
Years × Child Marriage				−0.018
				(0.013)
Rural		−0.338***	−0.202**	−0.207***
		(0.080)	(0.079)	(0.079)
State FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Interaction	No	No	No	Yes
Currently Married		0.041	0.060	0.050
		(0.082)	(0.081)	(0.081)
Observations	932	932	932	932
R^2	0.459	0.470	0.539	0.540
Adj. R^2	0.457	0.467	0.527	0.527

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied.

Table 11: Returns by Education Level – N. Nigeria

Table 15: Returns by Education Level – N. Nigeria

	By Level
Primary	−0.319
	(0.351)
Secondary	0.215
	(0.150)
Higher Ed	1.912***
	(0.139)
Experience	0.044***
	(0.004)
Experience Sq.	−0.002***
	(0.000)
Child Marriage	−0.190*
	(0.105)
N	932
R2	0.521

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note:
Omitted: No education. State FE, rural, marital controls included.

Table 16: Returns to Literacy – N. Nigeria

	Literacy
Literate	1.529***
	(0.119)
Experience	0.035***
	(0.004)
Experience Sq.	−0.002***
	(0.000)
Child Marriage	−0.309***
	(0.110)
N	932
R2	0.461

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred.

Table 17: Probit: Pr(Wage Employment) – N. Nigeria

	Pr(Wage Work)
Years of Schooling	0.047***
	(0.005)
Years Sq.	0.009***
	(0.001)
Experience	0.026***
	(0.002)
Experience Sq.	−0.001***
	(0.000)
Child Marriage	−0.094*
	(0.050)
Rural	−0.111**
	(0.044)
Observations	16 140
State FE	Yes
Parent Education	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE.

Table 14: Average Marginal Effects – N. Nigeria

Table 18: Average Marginal Effects on Pr(Wage Work) – N. Nigeria

Variable	AME	SE	p
Child Marriage	-0.0080	0.0042	0.056
Experience	0.0023	0.0002	< 0.001
Experience Sq.	-0.0001	0.0000	< 0.001
Rural	-0.0099	0.0041	0.015
Years of Schooling	0.0041	0.0004	< 0.001
Years of Schooling Sq.	0.0008	0.0001	< 0.001

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. AME via marginaleffects package.

Table 15: Causal Forest – N. Nigeria

Average Return to Education

Table 19: CF: Average Treatment Effect – N. Nigeria

Term	Estimate	Std. Error	t stat	p value
Education Return (per year)	0.149	0.016	9.389	0

Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees.

Test for Heterogeneity

Table 20: CF Calibration Test – N. Nigeria

Term	Estimate	Std. Error	t stat	p value
Mean Forest Prediction	1.055	0.163	6.461	0.000
Differential Forest Prediction	0.969	0.432	2.243	0.013

Variable Importance

Table 21: CF Variable Importance – N. Nigeria

Variable	Importance
Experience	0.380
Age	0.250
Mother's Education	0.161
Father's Education	0.126
Child Marriage	0.046
Rural	0.037

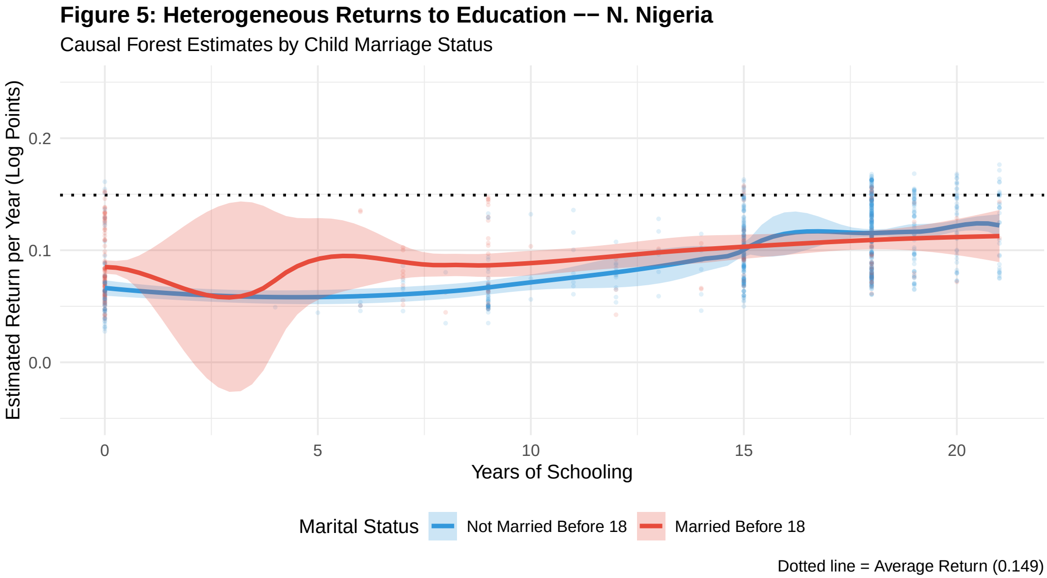


Figure 5: Heterogeneous Returns to Education –– N. Nigeria
Causal Forest Estimates by Child Marriage Status
Estimated Return per Year (Log Points)
0.2
0.1
0.0
0
5
10
15
20
Years of Schooling
Marital Status
Not Married Before 18
Married Before 18
Dotted line = Average Return (0.149)

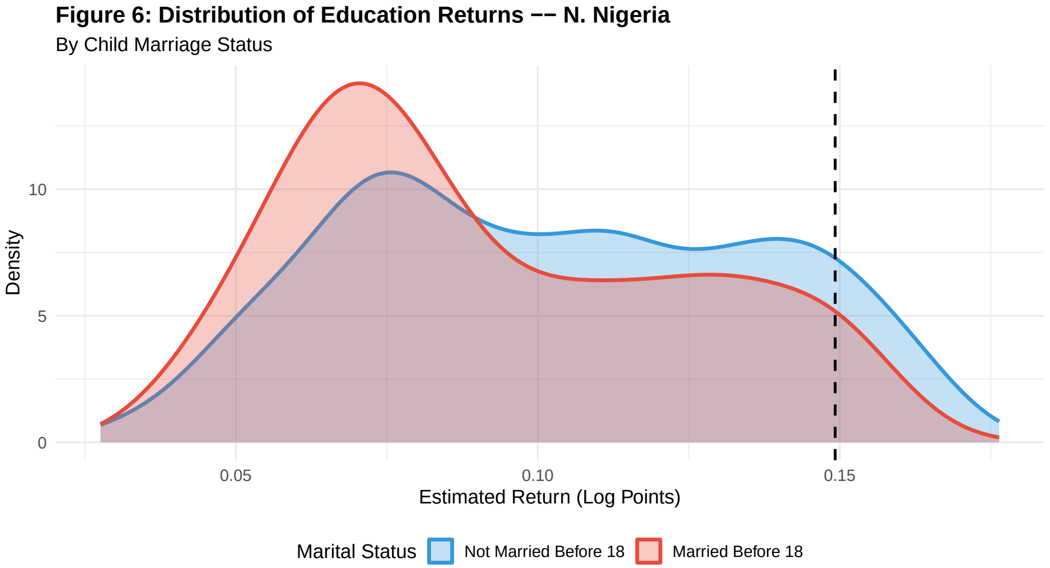


Figure 6: Distribution of Education Returns –– N. Nigeria
By Child Marriage Status
Density
10
5
0
0.05
0.10
0.15
Estimated Return (Log Points)
Marital Status
Not Married Before 18
Married Before 18

Figure 7: Returns by Level and Marriage – N. Nigeria

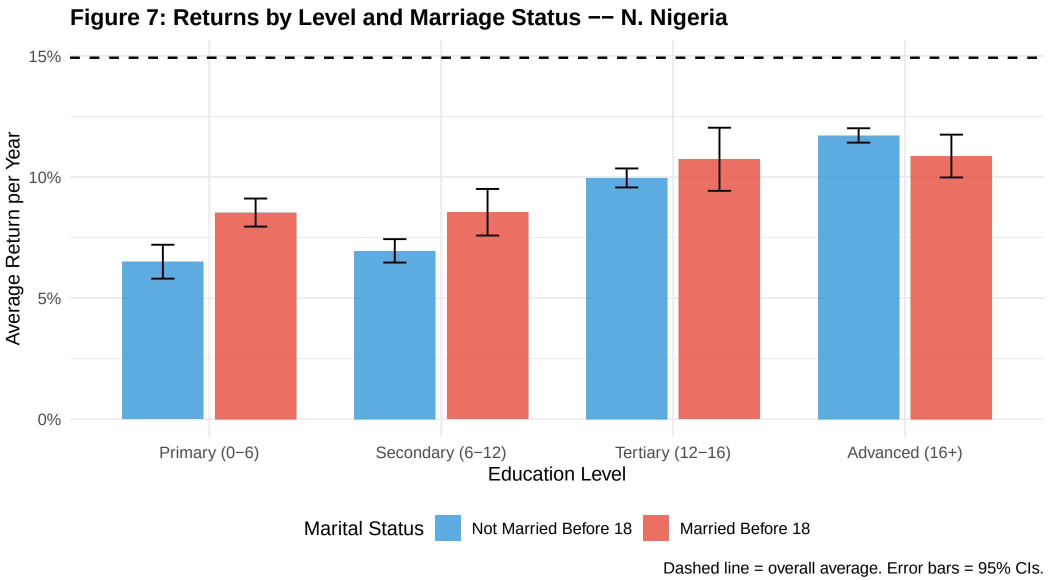
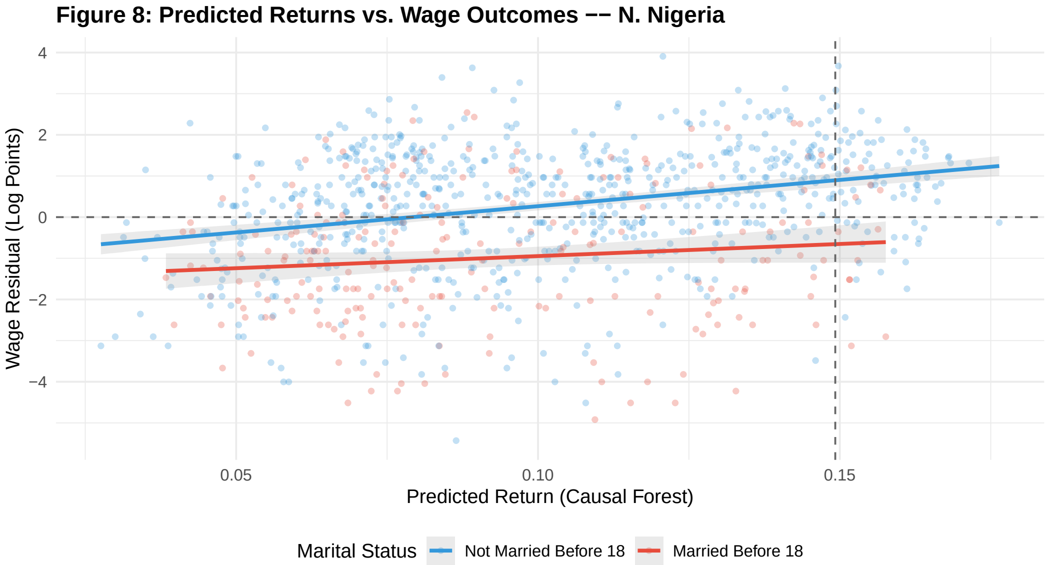


Figure 8: Validation – N. Nigeria



Cross-Region Comparison

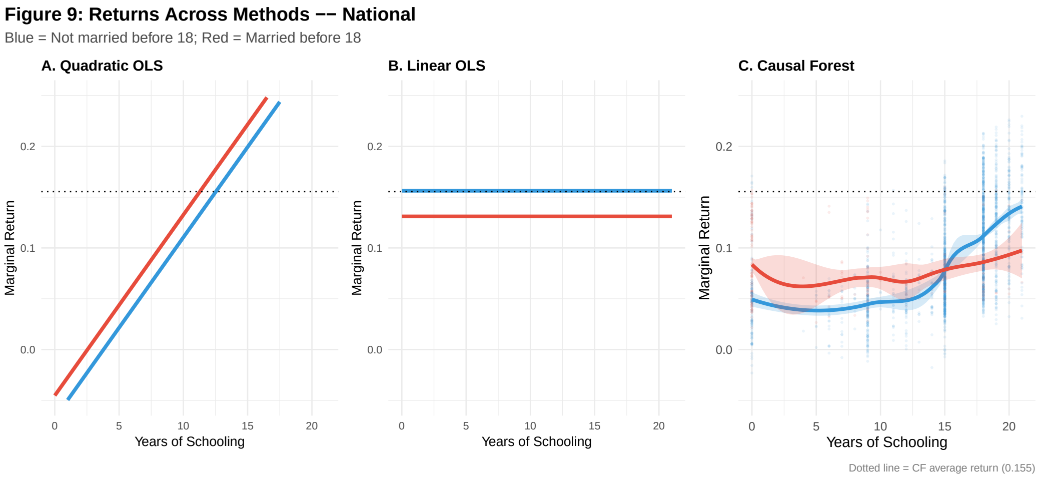


Figure 9: Returns Across Methods –– National
Blue = Not married before 18; Red = Married before 18
A. Quadratic OLS
B. Linear OLS
C. Causal Forest
Marginal Return
0.2
0.1
0.0
0
5
10
15
20
Years of Schooling
Dotted line = CF average return (0.155)

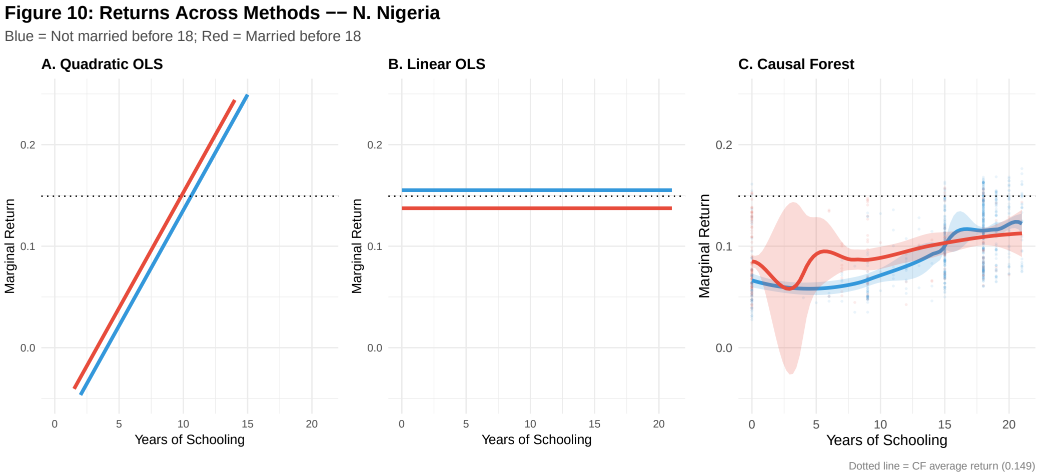


Figure 10: Returns Across Methods –– N. Nigeria
Blue = Not married before 18; Red = Married before 18
A. Quadratic OLS
B. Linear OLS
C. Causal Forest
Marginal Return
0.2
0.1
0.0
0
5
10
15
20
Years of Schooling
Dotted line = CF average return (0.149)

Table 16: Key Estimates Comparison

Table 22: Key Estimates – National vs. Northern Nigeria

Metric	National	N. Nigeria
Average Return (CF)	15.5%	14.9%
Return Std. Error	0.017	0.016
Child Marriage Penalty (OLS+FE)	-7.8%	-15.0%
Interaction: Years x Child Marriage	0.022	0.017
Interaction p-value	0.054	0.155
Child Marriage Var. Importance (CF)	6.4%	4.6%
Heckman lambda	0.822**	0.217 (n.s.)
N (wage earners)	2,387	932
N (selection sample)	26,450	16,140

Note: Source: NLSS 2018-19. CF = Causal Forest. OLS+FE = OLS with state fixed effects.

Appendix

Table 23: Descriptive Statistics – Full Working-Age Sample

Variable	National (N=26,450)	N. Nigeria (N=16,140)
Continuous: Mean (SD)		
Log annual wage	12.02 (1.42)	11.83 (1.60)
Years of schooling	8.37 (7.07)	5.90 (6.92)
Age	34.57 (12.26)	33.50 (11.69)
Experience	18.90 (14.26)	19.58 (13.69)
Binary (%)		
Child marriage (%)	31.6	46.2
Rural (%)	70.6	77.9
Literate (%)	59.7	47.5
Currently married (%)	69.3	78.4
Religion (%)		
Religion: Islam (%)	51.6	28.3
Religion: Christianity (%)	47.8	71.2
Religion: Other (%)	0.7	0.5
Father education (%)		
Father edu: None (%)	40.1	40.2
Father edu: Primary (%)	29.1	16.0
Father edu: Secondary+ (%)	30.8	43.8
Mother education (%)		
Mother edu: None (%)	50.6	51.3
Mother edu: Primary (%)	25.3	13.1
Mother edu: Secondary+ (%)	24.1	35.5
Sector (%)		
Sector: Urban (%)	29.4	22.1
Sector: Rural (%)	70.6	77.9
Labour market (%)		
Wage employment (%)	8.7	5.6
N	26,450	16,140